

AI PIPELINE V1

POST api/v1/ai/simplify

input:

```
{
  "raw_text": "Comprehensive Metabolic Panel & CBC:\nWBC: 5.4 x10^3/uL (Normal: 4.0 - 11.0)\nHgb: 11.1 g/dL (Low: 13.0 - 17.0)\nMCV: 74 fL (Low: 80 - 100)\nPlatelets: 285 x10^3/uL (Normal: 150 - 450)\n\nMetabolic Profile:\nFasting Blood Glucose: 118 mg/dL (High, Normal: 70 - 99)\nHbA1c: 6.2% (Prediabetes: 5.7 - 6.4%)\nSerum Creatinine: 0.9 mg/dL (Normal: 0.7 - 1.3)\neGFR: >90 mL/min (Normal: >60)\n\nImpression: Microcytic anemia and impaired fasting glucose consistent with prediabetes. Recommend nutritional consultation and iron evaluation."
}
```

output and gemini recommendation:

Checking the server response body from your Swagger UI for **POST /api/v1/ai/simplify:**

JSON

```
{
  "title": "Fasting Blood Glucose & Diabetes Profile Analysis",
  "raw_text": "Comprehensive Metabolic Panel & CBC:\nWBC: 5.4 x10^3/uL (Normal: 4.0 - 11.0)\nHgb: 11.1 g/dL (Low: 13.0 - 17.0)\nMCV: 74 fL (Low: 80 - 100)\nPlatelets: 285 x10^3/uL (Normal: 150 - 450)\n\nMetabolic Profile:\nFasting Blood Glucose: 118 mg/dL (High, Normal: 70 - 99)\nHbA1c: 6.2% (Prediabetes: 5.7 - 6.4%)\nSerum Creatinine: 0.9 mg/dL (Normal: 0.7 - 1.3)\neGFR: >90 mL/min (Normal: >60)\n\nImpression: Microcytic anemia and impaired fasting glucose consistent with prediabetes. Recommend nutritional consultation and iron evaluation.",
  "extracted_entities": [
    {
      "entity": "Platelet Count",
      "value": "285 /uL",
      "reference": "150,000 - 450,000 /uL",
      "status": "Low"
    },
    {
      "entity": "WBC Count",

```

```
"value": "5.4 x10^3/uL",  
"reference": "4.0 - 11.0 x10^3/uL",  
"status": "Normal"
```

```
},
```

```
{
```

```
"entity": "Hemoglobin",  
"value": "11.1 g/dL",  
"reference": "13.0 - 17.0 g/dL",  
"status": "Low"
```

```
},
```

```
{
```

```
"entity": "HbA1c",  
"value": "6.2 %",  
"reference": "< 5.7 %",  
"status": "High"
```

```
},
```

```
{
```

```
"entity": "Fasting Blood Sugar",  
"value": "118.0 mg/dL",  
"reference": "70 - 99 mg/dL",  
"status": "High"
```

```
}
```

```
],
```

```
"standard_summary": "Your Complete Blood Count shows abnormal levels for: Hemoglobin  
(11.1 g/dL). Your Blood Sugar test results indicate elevated glucose levels: HbA1c (6.2 %),  
Fasting Blood Sugar (118.0 mg/dL).",
```

```
"eli5_summary": "Imagine your bloodstream is like a smooth highway. Too much sugar is  
like traffic slowing down the cars, and high cholesterol is like sticky oil on the road. Eating
```

less sweets, drinking plenty of water, taking your prescribed medicine, and going for daily 30-minute brisk walks will clear up the highway so your body runs fast and smooth!",

"recommendations": [

"Switch to a diabetic-friendly low glycemic index diet.",

"Take prescribed oral hypoglycemic medications as directed by your physician.",

"Monitor fasting and post-prandial blood sugar levels regularly."

],

"is_flagged": true,

"flagged_reason": "Platelet Count 285 /uL (LOW) | Hemoglobin 11.1 g/dL (LOW) | HbA1c 6.2 % (HIGH) | Fasting Blood Sugar 118.0 mg/dL (HIGH)"

}

Verification: Partially Incorrect / 3 Specific Bugs Detected

1. Unit Normalization & False Positive on Platelet Count

- **Input:** Platelets: 285 x10³/uL (Normal: 150 - 450) (which is 285,000 /μL, completely normal).
- **Extracted Entity:** "value": "285 /uL", with reference "150,000 - 450,000 /uL", marking it "**Low**" and adding it to flagged_reason.
- **Issue:** The parsing logic dropped the scientific multiplier x10³ (or thousand multiplier) from the value while comparing it against an unscaled reference range, causing a normal platelet count to be falsely flagged as dangerously low.

2. Hallucinated Cholesterol Reference in eli5_summary

- The ELI5 summary states: *"...and high cholesterol is like sticky oil on the road."*
- **Issue:** The input report is a CMP + CBC (anemia and blood glucose); it does **not** contain a lipid panel or cholesterol tests. The model carried over an analogy from an earlier prompt or few-shot example.

3. Missed Entities from the Input

- **MCV (74 fL - Low):** Missed, even though the clinical impression explicitly calls out *microcytic anemia*.
- **Creatinine (0.9 mg/dL) & eGFR (>90 mL/min):** Missed completely from the metabolic panel extraction.

Suggested Fixes for the Pipeline

1. **Entity Extractor:** Normalize unit notations (e.g., $\times 10^3/\text{uL}$, k/uL , and absolute counts) before evaluating high/low status.
2. **Prompt Boundary / Guardrail:** Instruct the LLM in the ELI5 generator to strictly use metaphors matching *only* the flagged conditions present in the extracted entities.

Medical reports

post/api/reports/uploads

input:

Form Field Values

Field	Type	Recommended Test Value
patient_id	integer	1
report_type	string	lab_report (or blood_test)
language	string	en
file	file / binary	Leave blank (or keep the checkbox checked for "Send empty value" if testing with text only)
raw_text	string	(Paste the report text below)

Sample Value for raw_text

Plaintext

PATIENT LAB REPORT:

Patient ID: 1

Date: 2026-09-19

Complete Blood Count (CBC):

- White Blood Cells (WBC): $6.8 \times 10^3/\text{uL}$ (Normal: 4.0 - 11.0)
- Red Blood Cells (RBC): $4.65 \times 10^6/\text{uL}$ (Normal: 4.20 - 5.80)
- Hemoglobin (Hgb): 14.2 g/dL (Normal: 13.0 - 17.0)
- Hematocrit (Hct): 42.1% (Normal: 38.0 - 50.0)

Lipid Panel:

- Total Cholesterol: 245 mg/dL (High, Desirable: <200)
- Triglycerides: 185 mg/dL (Borderline High: 150-199)
- HDL Cholesterol: 38 mg/dL (Low, Desirable: >40)
- LDL Calculated: 170 mg/dL (High, Optimal: <100)

Impression: Hyperlipidemia with elevated LDL and reduced HDL. Recommend dietary intervention and regular aerobic exercise.

output and recommendation:

The upload endpoint **failed** with an HTTP **500 Internal Server Error**.

Executed Request & Server Response

- **Request URL:** http://localhost:8000/api/reports/upload
- **Status Code:** 500 Undocumented (Error: Internal Server Error)
- **Response Body:**

Plaintext

Internal Server Error

- **Payload Sent:**
 - patient_id: 1
 - report_type: lab_report
 - raw_text: string
 - language: en
 - file: @PATIENT LAB REPORT.txt;type=text/plain

Why the 500 Error Occurred (Common Root Causes)

1. **Foreign Key Constraint / Missing Patient:**
 - The request sent patient_id: 1. If user/patient ID 1 does not exist in the database (or if authentication is required to link records), the database query or insertion throws an unhandled IntegrityError / ForeignKeyViolation.
2. **File Handling / Storage Directory Missing:**

- When uploading PATIENT LAB REPORT.txt, the backend tries to save the file locally (e.g., to an uploads/ directory) or parse it. If the upload folder does not exist or lacks write permissions, an unhandled FileNotFoundError or PermissionError is thrown.

3. File Parser / OCR Exception:

- If the upload handler attempts to pass the uploaded file through an OCR/parsing library (e.g., PyPDF2, pdf2image, tesseract, or text reader) without handling plain text files properly, an unhandled exception occurs.

How to Diagnose & Fix in Backend

- **Check the Uvicorn terminal console:** Inspect the traceback to pinpoint the exact line causing the crash.
- **Verify Patient Exists:** Make sure you register a user/patient first (via POST /api/auth/register) and pass their valid patient_id.
- **Add Try/Except:** Wrap the file storage and extraction logic in a try...except block so it returns a clean 400 Bad Request or 422 Unprocessable Entity instead of an unhandled 500.

Yes, they will either return errors or fail to return meaningful data, depending on how each endpoint is implemented:

1. GET /api/reports/{report_id}

- **Will Fail (404 or 500):** Since no report was successfully created or saved to the database, querying any specific ID (e.g., /reports/1) will return:
 - **404 Not Found** (if properly handled when a record does not exist).
 - **500 Internal Server Error** (if the code assumes the record exists and attempts to access properties on None).

2. GET /api/reports/compare/trends

- **Will Fail (400, 404, or 500):**
 - Biomarker trend comparisons require at least one or two historical reports to compute trajectories and trend lines.
 - Without any stored reports in the database, the comparison algorithm will typically return a **400 Bad Request** ("Insufficient reports for trend analysis"), a **404 Not Found**, or crash with a **500 Internal Server Error** (e.g., trying to calculate deltas on empty data structures).

3. GET /api/reports/

- **Might return 200 with an empty list ([]), OR throw 401/500:**
 - If the database table exists and the query executes cleanly, it will return **200 OK** with an empty list: [].
 - However, if the endpoint requires an authenticated user session/token that hasn't been provided, it will return **401 Unauthorized**.
 - If the table itself has not been migrated or created in your database, it will return **500 Internal Server Error**.

Recommended Next Step

Check your backend terminal (where Uvicorn is running) to inspect the traceback for the 500 error on /upload. Once the upload handler writes successfully to the database, all downstream GET and /compare/trends endpoints will have valid data to query and display.